

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Searches for and finds correct counter-example to the associativity condition	AO1.1b	M1	$(3\Delta 1)\Delta 2 = -2$ $3\Delta(1\Delta 2) = -4$
Correctly argues that the binary operation is not associative, disproving Gary's claim	AO2.3	R1	$(3\Delta 1)\Delta 2 \neq 3\Delta(1\Delta 2)$ Therefore Δ is not an associative binary operation, which disproves Gary's claim.
Total 2 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows that the set is closed under the operation $*$ (must show that under modulo 6, $a * b$ can only result in a member of the given set)	AO2.1	R1	As all answers to $a * b$ are reduced modulo 6 they are in the given set and thus the set is closed under $*$.
	Clearly identifies the identity element	AO1.1b	B1	Identity element = 2
	Finds and states the inverse of each element (PI) FT from 'their' identity	AO1.1b	B1	0 and 4 are inverses of each other 1 and 3 are inverses of each other 2 and 5 are self-inverse elements
	Shows associativity between elements of the set under the operation $*$	AO2.1	R1	$a * (b * c) = a + (b + c + 4) + 4$ is shown to equal $(a * b) * c = (a + b + 4) + c + 4$
	States correct conclusion that G is a group under the operation $*$ Only award if they have a	AO2.1	R1	As G satisfies each of the four group axioms under the binary operation $*$, G is a group

	completely correct solution, which is clear, easy to follow and contains no slips.																	
(b)	Identifies two correct subgroups. Condone inclusion of {0,1,2,3,4,5}	AO1.1b	B1	{2}, {0, 2, 4}, {2, 5}														
	Identifies all three proper subgroups and no others included	AO1.1b	B1															
(c)	Identifies the generator of G OR generates every element of the group K (PI)	AO3.1a	B1	$G = (\langle 1 \rangle, *)$ OR $K = (\{3,9,13,11,5,1\}, \times_{14})$														
	Finds correctly a one-to-one mapping between each element of G and K (condone use of equal sign for this mark)	AO1.1b	B1	<table><thead><tr><th>G</th><th>K</th></tr></thead><tbody><tr><td>1</td><td>$\mapsto$ 3</td></tr><tr><td>0</td><td>$\mapsto$ 9</td></tr><tr><td>5</td><td>$\mapsto$ 13</td></tr><tr><td>4</td><td>$\mapsto$ 11</td></tr><tr><td>3</td><td>$\mapsto$ 5</td></tr><tr><td>2</td><td>$\mapsto$ 1</td></tr></tbody></table>	G	K	1	$\mapsto$ 3	0	$\mapsto$ 9	5	$\mapsto$ 13	4	$\mapsto$ 11	3	$\mapsto$ 5	2	$\mapsto$ 1
	G	K																
1	$\mapsto$ 3																	
0	$\mapsto$ 9																	
5	$\mapsto$ 13																	
4	$\mapsto$ 11																	
3	$\mapsto$ 5																	
2	$\mapsto$ 1																	
	Deduces that G is isomorphic to K with a concluding statement using the correct mathematical language, having used the correct notation throughout.	AO2.2a	E1	As there is a one-to-one mapping between the elements of G and the elements of K , $G \cong K$														
Total 10 marks																		